

Derivation of the Wasserstein Gradient Flow Equation

In Euclidean space, the gradient flow of a function f is

$$\dot{x}_t = -\nabla f(x_t).$$

This says that the point moves in the direction of steepest decrease, measured using the Euclidean inner product.

In Wasserstein space, points are probability densities ρ_t . A curve of densities moves through a velocity field v_t satisfying the continuity equation:

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0.$$

The Wasserstein metric measures the size of the velocity by kinetic energy:

$$\|v_t\|_{\rho_t}^2 = \int_{\mathbb{R}^d} |v_t(x)|^2 \rho_t(x) dx.$$

Derivative of the Functional

Let $\mathcal{F}$ be a functional on densities. Along a curve ρ_t ,

$$\frac{d}{dt} \mathcal{F}(\rho_t) = \int_{\mathbb{R}^d} \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t)(x) \partial_t \rho_t(x) dx.$$

Using the continuity equation,

$$\frac{d}{dt} \mathcal{F}(\rho_t) = - \int_{\mathbb{R}^d} \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t) \operatorname{div}(\rho_t v_t) dx.$$

Integrating by parts gives

$$\frac{d}{dt} \mathcal{F}(\rho_t) = \int_{\mathbb{R}^d} \nabla \left[\frac{\delta \mathcal{F}}{\delta \rho}(\rho_t) \right] \cdot v_t \rho_t dx.$$

Identifying the Wasserstein Gradient

The Wasserstein inner product of two velocity fields is

$$\langle u, v \rangle_{\rho_t} = \int_{\mathbb{R}^d} u \cdot v \rho_t dx.$$

The previous identity shows that the velocity-field representative of the Wasserstein gradient of $\mathcal{F}$ is

$$\nabla \left[\frac{\delta \mathcal{F}}{\delta \rho}(\rho_t) \right].$$

Therefore the negative-gradient velocity is

$$v_t = -\nabla \left[\frac{\delta \mathcal{F}}{\delta \rho}(\rho_t) \right].$$

Plugging Into the Continuity Equation

Substituting this velocity into

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0$$

gives

$$\partial_t \rho_t = \operatorname{div} \left(\rho_t \nabla \left[\frac{\delta \mathcal{F}}{\delta \rho}(\rho_t) \right] \right).$$

This is the Wasserstein gradient flow equation.